MXB103 Cheat Sheet Side 1: Q1–Q8

Calculator in **RADIANS**; log = ln. Show working: the formula line usually earns 1 mark.

W1 Floating point Q1

$F(\beta,k,m,M)$: $\pm d_1.d_2\cdots d_k\times\beta^e$, $d_1\neq 0$, $m\le e\le M$ (k counts *all* digits)

- $S=\beta^m$ $L=(\beta-\beta^{1-k})\beta^M$ $u=\frac12\beta^{1-k}$
- abs $=|x-\tilde x|$, rel $=\frac{|x-\tilde x|}{|x|}$ (divide by the **true** value)
- dec→bin: integer ÷2, read remainders **up**; fraction ×2, read integer parts **down**
- $fl(x)$: base β → normalise → check e → round to k digits (recheck e if rounding carries)
- $|x|<S$: underflow → 0; $|x|>L$: overflow → Inf. IEEE double $F(2,53,-1022,1023)$, $u\approx 1.11\times10^{-16}$

e.g. $24.25=11000.01_2$; $10001.001_2=17.125$; $F(2,5,-4,4)$: $S=2^{-4}=0.0625$; $F(3,6,\cdot,\cdot)$: $u=\frac12 3^{-5}=0.0021$

W2 Cancellation & Taylor Q1b, Q2

Cancellation: subtracting nearly-equal *rounded* numbers. Fix: multiply by the conjugate, or use log/trig rules.

- $\frac{-b+\sqrt{b^2-4ac}}{2a}=\frac{-2c}{b+\sqrt{b^2-4ac}}$ ($b>0$); for $b<0$ the other root is $\frac{-2c}{b-\sqrt{\cdot}}$
- $1-\cos x=\frac{\sin^2 x}{1+\cos x}$; $\log(x+1)-\log x=\log(1+\frac1x)$; $\sqrt{x+1}-\sqrt{x}=\frac{1}{\sqrt{x+1}+\sqrt{x}}$

Taylor: $P_n(x)=\sum_{k=0}^n\frac{f^{(k)}(x_0)}{k!}(x-x_0)^k$

$$R_n(x)=\frac{f^{(n+1)}(c)}{(n+1)!}(x-x_0)^{n+1}$$

with c between x_0 and x. **Bound:**

$$|f-P_n|\le\frac{\max|f^{(n+1)}|}{(n+1)!}\cdot\max|x-x_0|^{n+1}$$

- max of $|f^{(n+1)}|$: check the interior too (e.g. $|\sin|$ on $[0,\pi]$ is 1)
- distance is from x_0: $x_0=\frac\pi6$ on $[0,\frac\pi3]\Rightarrow\frac\pi6$
- marks: form of R_n [1], working [1], number [1]; state where c lies

$\cos x$, $n=2$, $x_0=0$: $P_2=1-\frac{x^2}{2}$, $R_2=\frac{\sin c}{6}x^3\le\frac{\sqrt3/2}{6}(\frac\pi3)^3=0.1658$ on $[0,\frac\pi3]$

W3–4 ODEs $y'=f(t,y)$, $y(a)=\alpha$ Q3, Q4, Q5

$h=\frac{b-a}{n}$, $t_i=a+ih$, $w_i\approx y_i=y(t_i)$. The output is a *table*, not a formula.

- **Euler:** $w_{i+1}=w_i+hf(t_i,w_i)$
- **Taylor 2:** $w_{i+1}=w_i+hf+\frac{h^2}{2}f'(t_i,w_i)$
- **!** $f'=f_t+f_y\cdot f$ (chain rule: y depends on t). $f=3t-\frac yt\Rightarrow f'=\frac{2y}{t^2}$
- **Mod. Euler:** $k_1=hf(t_i,w_i)$, $k_2=hf(t_i+h,w_i+k_1)$, $w_{i+1}=w_i+\frac12(k_1+k_2)$
- **RK4:** $k_1=hf(t_i,w_i)$, $k_2=hf(t_i+\frac h2,w_i+\frac{k_1}{2})$, $k_3=hf(t_i+\frac h2,w_i+\frac{k_2}{2})$, $k_4=hf(t_i+h,w_i+\mathbf{k_3})$; $w_{i+1}=w_i+\frac16(k_1+2k_2+2k_3+k_4)$

	Euler	Tay. 2	M. Euler	RK4
global	$O(h)$	$O(h^2)$	$O(h^2)$	$O(h^4)$
f'?	no	**yes**	no	no

Local $O(h^{p+1})\Rightarrow$ global $O(h^p)$ ($\frac{b-a}{h}$ steps). Halve $h\Rightarrow$ error $\div 2^p$.

Accuracy template: "(b) is more accurate: [method] is **second order** $O(h^2)$, Euler only **first order** $O(h)$."

Derive (Q5), 4 moves: **1** Taylor $y(t)$ about t_i (degree 1 or 2) **2** put $t=t_{i+1}$, $t_{i+1}-t_i=h$ **3** $y'=f$, $y''=f'$ **4** drop the $O(\cdot)$ term, $y\to w$.

$y_{i+1}=y_i+hf_i+\frac{h^2}{2}f_i'+O(h^3)$ (Taylor 2)

Mod. Euler, extra steps [0.5 each for moves 1–3, then]:

- [0.5] $f'(t_i,y_i)=\frac{f(t_{i+1},y_{i+1})-f(t_i,y_i)}{h}+O(h)$
- [1] substitute: $y_{i+1}=y_i+\frac h2[f(t_i,y_i)+f(t_{i+1},y_{i+1})]+O(h^3)$
- [1] w_{i+1} is on both sides ⇒ inside f use Euler $w_{i+1}\approx w_i+hf(t_i,w_i)$

$$w_{i+1}=w_i+\tfrac h2\big[f(t_i,w_i)+f(t_{i+1},w_i+hf(t_i,w_i))\big]$$

MATLAB (Q4): `t`, `w` initialised to a,α? `h=(b-a)/n`? `f(t,w)` not `f(w,t)`? `fdash` has `*f(t,y)`? RK4: $\frac k2$, full k_3, `/6`? loop `1:n`? `t=t+h` after the update?

W5 Lagrange Q6

$n+1$ distinct points ⇒ a **unique** P_n of degree $\le n$ (all forms give the same one).

$$P_n=\sum_i L_{n,i}(x)y_i,\quad L_{n,i}=\prod_{j\neq i}\frac{x-x_j}{x_i-x_j}$$

$$P_2=\frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)}y_0+\frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)}y_1+\frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)}y_2$$

- $L_{n,i}=1$ at x_i, 0 at the other nodes; **check** $\sum_i L_{n,i}(x)=1$
- $f-P_n=\frac{f^{(n+1)}(c)}{(n+1)!}(x-x_0)\cdots(x-x_n)$: bound $=\frac{\max|f^{(n+1)}|}{(n+1)!}\times|\prod|$ (evaluate at the point, or use the max over the interval, usually given)
- **MATLAB:** `.*` on vectors; denominator L_i uses (x_i-x_j), $j\neq i$; `rel=abs(P-f(a))/abs(f(a))`; `x(1)` is x_0

W6 Divided differences Q8

$f[x_i]=y_i$, $f[x_i..x_{i+k}]=\dfrac{f[x_{i+1}..x_{i+k}]-f[x_i..x_{i+k-1}]}{x_{i+k}-x_i}$ (last x − first x of the span)

$$P_n=f[x_0]+f[x_0,x_1](x-x_0)+f[x_0,x_1,x_2](x-x_0)(x-x_1)+\cdots$$

Coefficients = the **top diagonal**. Adding a point = 1 row + 1 term.

Q8 (1 mark/line): apply the definition repeatedly:

$$f[x_0..x_3]=\frac{f[x_1,x_2,x_3]-f[x_0,x_1,x_2]}{x_3-x_0}$$
$$f[x_1,x_2,x_3]=\frac{f[x_2,x_3]-f[x_1,x_2]}{x_3-x_1}$$
$$f[x_0,x_1,x_2]=\frac{f[x_1,x_2]-f[x_0,x_1]}{x_2-x_0}$$
$$f[x_i,x_{i+1}]=\frac{y_{i+1}-y_i}{x_{i+1}-x_i}$$

then substitute everything into line 1 (4 lines = 4 marks).

W7 Forward differences (equal h) Q7, Q8

$\Delta y_i=y_{i+1}-y_i$; $\Delta^2y_i=y_{i+2}-2y_{i+1}+y_i$; $\Delta^3y_i=y_{i+3}-3y_{i+2}+3y_{i+1}-y_i$

$$P_n=y_0+s\Delta y_0+\frac{s(s-1)}{2!}\Delta^2y_0+\frac{s(s-1)(s-2)}{3!}\Delta^3y_0,\qquad s=\frac{x-x_0}{h}$$

- **!** convert $x\to s$ first, then evaluate. Marks: table 1.5, form 0.5, polynomial 0.5, s 0.5, value of s 0.5, answer 0.5
- $f[x_0..x_k]=\dfrac{\Delta^k y_0}{k!\,h^k}$. Proof $k=2$: $\frac{\frac{y_2-y_1}{h}-\frac{y_1-y_0}{h}}{2h}=\frac{y_2-2y_1+y_0}{2h^2}$
- inside $[x_0,x_n]$ reliable; **extrapolation** ($s<0$ or $s>n$) unreliable

$(0,0),(\frac\pi4,.7071),(\frac\pi2,1)$: Δ: .7071, .2929; $\Delta^2=-.4142$; $s=\frac{4x}{\pi}$; $\sin\frac\pi6\approx P(s{=}\frac23)=0.5174$

Common traps, side 1

- Euler uses the slope at the **start** (t_i,w_i); update t after w
- one step of Mod. Euler: $y'=3+ty^3$, $y(2)=1$, $h=0.1$: $k_1=0.5$, $k_2=0.1f(2.1,1.5)=1.00875$, $w_1=1.754375$
- Taylor 2 derivative without the chain rule earns **0**
- error bound: use the $(n+1)$th derivative and the distance from x_0; state where c lies
- Lagrange: nodes distinct; numerator skips its own node; $\sum L=1$
- forward differences: equal spacing only; evaluate with s, not x

MXB103 Cheat Sheet Side 2: Q9–Q16

W8 Bisection & fixed point Q9, Q10

Bracket (2 marks): "$f(a) = \cdots < 0$, $f(b) = \cdots > 0$: opposite signs and f continuous, so by the IVT $[a, b]$ brackets a root."

- **Bisection:** $p = \frac{a+b}{2}$; if sign $f(p) =$ sign $f(a)$ then $a \leftarrow p$, else $b \leftarrow p$. Estimate after n iterations = the nth midpoint; $|p_n - r| \le \frac{b-a}{2^n}$
- table columns: iter | a | b | p | $f(a)$ | $f(b)$ | $f(p)$ (signs)
- **Fixed point:** $x = g(x)$, $x_{n+1} = g(x_n)$. Converges if $g : [a, b] \to [a, b]$ and $|g'| \le k < 1$ (sufficient, not necessary). Linear, rate $\lambda \approx |g'(p)|$
- **MATLAB:** `p=(a+b)/2`; else-branch `b=p`; `break` when `(b-a)<=tol`

W9 Newton & secant Q9, Q10, Q11

$$\textbf{Newton: } x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Secant:

$$x_{n+1} = x_n - f(x_n)\frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})}$$

Derive Newton [1,1,1]: tangent $f(x) \approx f(x_n) + f'(x_n)(x - x_n)$; set $0 = f(x_n) + f'(x_n)(x_{n+1} - x_n)$; solve.

Orders: bisection 1 ($\times\frac{1}{2}$), fixed point 1, secant $\frac{1+\sqrt{5}}{2} \approx 1.618$, Newton 2 (digits double). Newton fails if $f'(x_n) = 0$; it can cycle; it is only linear at a repeated root.

Q11 recipe: $f = x^k - a$, apply the method, simplify (use $p^2 - q^2 = (p - q)(p + q)$, $p^3 - q^3 = (p - q)(p^2 + pq + q^2)$).

- Newton $\sqrt{a}$: $\frac{1}{2}(x_n + \frac{a}{x_n})$ Newton $\sqrt[3]{a}$: $\frac{2x_n}{3} + \frac{a}{3x_n^2}$
- Secant $\sqrt{a}$: $x_n + \dfrac{a - x_n^2}{x_n + x_{n-1}}$ Secant $\sqrt[3]{a}$: $x_n - \dfrac{x_n^3 - a}{x_n^2 + x_n x_{n-1} + x_{n-1}^2}$
- **MATLAB:** `while abs(x-x_old) > tol`; `x - f(x)/fdash(x)`; secant: $f(x_1) - f(x_0)$ in the denominator, then shift `x0=x1; x1=x2`

W10 Numerical differentiation Q11, Q12

- fwd (left end): $f' \approx \dfrac{f(x_0 + h) - f(x_0)}{h}$, error $-\frac{h}{2}f''(c)$, $O(h)$
- bwd (right end): $f' \approx \dfrac{f(x_0) - f(x_0 - h)}{h}$, error $+\frac{h}{2}f''(c)$, $O(h)$
- central (middle): $f' \approx \dfrac{f(x_0 + h) - f(x_0 - h)}{2h}$, error $-\frac{h^2}{6}f'''(c)$, $O(h^2)$
- $f'' \approx \dfrac{f(x_0 - h) - 2f(x_0) + f(x_0 + h)}{h^2}$, error $-\frac{h^2}{12}f^{(4)}(c)$
- 2nd-order fwd: $f' \approx \dfrac{-3f(x_0) + 4f(x_0 + h) - f(x_0 + 2h)}{2h}$

Derive f'' (4 marks): [1] expand $f(x_0 \pm h) = f_0 \pm hf_0' + \frac{h^2}{2}f_0'' \pm \frac{h^3}{6}f_0''' + \frac{h^4}{24}f^{(4)}(c_{1,2})$; [1] **add** (odd terms cancel); [1] rearrange, $\div h^2$: error $-\frac{h^2}{12} \cdot \frac{f^{(4)}(c_1)+f^{(4)}(c_2)}{2}$; [1] **IVT**: the average $= f^{(4)}(c)$. (For f': **subtract**.)

Choice of h (2 marks): too large ⇒ **truncation** error; too small ⇒ **round-off** error (cancellation, then $\div$ small h). Best fwd $h \approx \sqrt{\varepsilon} \approx 10^{-8}$.

W11 Integration Q13, Q14

$$T = \tfrac{h}{2}\left[f_0 + 2(f_1 + \cdots + f_{n-1}) + f_n\right]$$

$$S = \tfrac{h}{3}\left[f_0 + 4(\text{odd}) + 2(\text{even int.}) + f_n\right]$$

Errors: T $-\frac{(b-a)h^2}{12}f''(c)$; S $-\frac{(b-a)h^4}{180}f^{(4)}(c)$.

- $n = 4$: T weights $1, 2, 2, 2, 1 \times \frac{h}{2}$; S weights $1, 4, 2, 4, 1 \times \frac{h}{3}$. Simpson needs **even** n
- T exact for degree ≤ 1, $O(h^2)$; S exact for degree ≤ 3, $O(h^4)$ ⇒ Simpson is more accurate
- **Derive T:** $P_1 = y_0 + s\Delta y_0$, $x = x_0 + sh$, $dx = h\,ds$: $\int_0^1 (y_0 + s\Delta y_0)h\,ds = h(y_0 + \frac{1}{2}\Delta y_0) = \frac{h}{2}(y_0 + y_1)$. Composite: sum the strips; interior points appear twice
- **Derive S:** $\int_0^2 [y_0 + s\Delta y_0 + (\frac{s^2}{2} - \frac{s}{2})\Delta^2 y_0]h\,ds = h[2y_0 + 2\Delta y_0 + \frac{1}{3}\Delta^2 y_0]$, then substitute the Δ's to get $\frac{h}{3}(y_0 + 4y_1 + y_2)$
- **MATLAB:** `h=(b-a)/n`; points `x(1)..x(n+1)`; `2*f` for interior; last point `x(n+1)`; Simpson `h/3`, even `n`; even MATLAB index `x(2)` is x_1, weight 4

W12 Triangular systems & LU Q15

- forward (lower): $x_i = \frac{b_i - \sum_{j<i} a_{ij}x_j}{a_{ii}}$ top→down; backward (upper): $x_i = \frac{b_i - \sum_{j>i} a_{ij}x_j}{a_{ii}}$ bottom→up
- **Explain LU (3 marks):** $A = LU \Rightarrow L(U\mathbf{x}) = \mathbf{b}$. Let $\mathbf{z} = U\mathbf{x}$: solve $L\mathbf{z} = \mathbf{b}$ (forward), then $U\mathbf{x} = \mathbf{z}$ (backward). Reuse L, U for a new $\mathbf{b}$
- **Fill blanks:** a_{ij} =(row i of L)·(col j of U); L has **1s on the diagonal**; one unknown per equation (order: row of U, column of L, …)
- $u_{1j} = a_{1j}$; $\ell_{i1} = \frac{a_{i1}}{u_{11}}$; $u_{22} = a_{22} - \ell_{21}u_{12}$; $u_{23} = a_{23} - \ell_{21}u_{13}$; $\ell_{32} = \frac{a_{32} - \ell_{31}u_{12}}{u_{22}}$; $u_{33} = a_{33} - \ell_{31}u_{13} - \ell_{32}u_{23}$
- a zero or small pivot breaks LU ⇒ pivoting (row swaps)

W13 SPD & Cholesky Q16

- **SPD:** $A^T = A$ and $\mathbf{x}^T A\mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$ (needed for Cholesky)
- **Show PD:** [1] state the goal; [1] $\mathbf{x}^T A\mathbf{x} = ax_1^2 + 2bx_1x_2 + cx_2^2$; [0.5] $= a(x_1 + \frac{b}{a}x_2)^2 + (c - \frac{b^2}{a})x_2^2$; [0.5] "$= 0$ only if $\mathbf{x} = \mathbf{0}$". PD $\iff$ $a > 0$, $ac - b^2 > 0$
- **Cholesky** $A = LL^T$; a_{ij} =(row i of L)·(row j of L): $\ell_{11} = \sqrt{a_{11}}$, $\ell_{i1} = \frac{a_{i1}}{\ell_{11}}$, $\ell_{22} = \sqrt{a_{22} - \ell_{21}^2}$, $\ell_{32} = \frac{a_{32} - \ell_{31}\ell_{21}}{\ell_{22}}$, $\ell_{33} = \sqrt{a_{33} - \ell_{31}^2 - \ell_{32}^2}$
- solve $L\mathbf{z} = \mathbf{b}$ (forward), then $L^T\mathbf{x} = \mathbf{z}$ (**backward**). Half the work of LU, no pivoting
- "Is A SPD?" → "Yes: symmetric, and the Cholesky factorisation succeeded." $\sqrt{\text{negative}}$ ⇒ not PD

MATLAB bug checklist (3 questions = 18 marks)

Write the correct algorithm first, then go line by line. For each bug: **line number**, **what's wrong**, **corrected line**.

- variable not initialised (`t`, `w`); wrong initial value
- argument order `f(t,w)`; `fdash` missing chain rule `*f(t,y)`
- `h=(b-a)/n`; midpoint `(a+b)/2`; coefficients ($k/2$, `2*`, `/6`, $h/3$)
- indices `x(1)`…`x(n+1)`; loop `1:n`; Simpson needs even `n`
- stop logic: `while …> tol`; `if …<= tol, break`
- formula upside down (`f/fdash`); update swapped (`b=p`, `x=xnew`)
- `.* ./ .^` on vectors; relative error ÷ `abs(true)`

Order of accuracy, all methods

- $O(h)$: Euler; forward/backward difference
- $O(h^2)$: Taylor 2, Mod. Euler; central f', f''; trapezoid
- $O(h^4)$: RK4; Simpson
- roots: bisection and fixed point 1; secant 1.618; Newton 2

Common traps, side 2

- bisection answer = the **latest midpoint**; keep the half with the sign change
- one secant iteration from x_0, x_1 gives x_2; write the general formula first (0.5)
- table end-points: forward at the left end, backward at the right end, central in the middle
- T uses $\frac{h}{2}$, S uses $\frac{h}{3}$; n counts strips (5 points means $n = 4$)
- LU: $\mathbf{z}$ is **not** the answer; solve $U\mathbf{x} = \mathbf{z}$ next
- Cholesky: the second solve with L^T goes **bottom-up**
- 2025 Q14: $T = 2.0959$, $S = 2.0934$ (check your calculator routine)